Highlights from the three-spring problem: the same problem, solved in two ways

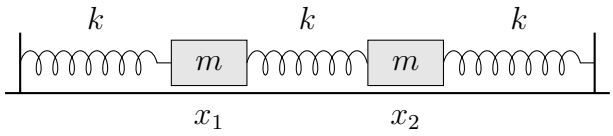
Michael G. Lerner

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These notes closely follow Felder and Felder, *Mathematical Methods in Engineering and Physics* (Wiley, 2016), Section 6.9, "Putting It Together: Revisiting the Three-Spring Problem" (pp. 336–340), sometimes to the point of plagiarism, so I'll just give full credit now. The book works the problem on a matrix with eigenvalues -4 and -1; here I've swapped in the carts we used in class.

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1 The problem



Question: Suppose we have two carts on a smooth track (perhaps an airtrack), each with mass m, and connected by three equal springs with spring constants k. Find the solution (i.e., $x_1(t)$ and $x_2(t)$) with initial conditions $x_1(0) = 3$ cm, $x_2(0) = 0$, $\dot{x}_1(0) = \dot{x}_2(0) = 0$. Here x_1 and x_2 are each cart's displacement from its own equilibrium position, with positive x to the right.

Step 1: draw the picture, write down N2L. Cart 1 feels the left spring, stretched by x_1, pulling it back, and the middle spring, stretched by $x_2 - x_1$, pulling it forward:

$$m\ddot{x}_1 = -kx_1 + k(x_2 - x_1) = -2kx_1 + kx_2 \tag{1.1}$$

$$m\ddot{x}_2 = -2kx_2 + kx_1 \tag{1.2}$$

(We get the second line by symmetry, just swapping the labels. Look at the signs: a cart's own displacement is opposed by both springs attached to it, which is the $-2kx_1$ in cart 1's equation; moving the *other* cart to the right stretches the middle spring, which pulls this cart to the right too, which is the $+kx_2$. In a moment those become the negative diagonal and positive off-diagonal entries of a matrix.)

Step 2: turn it into a matrix equation. Writing $\omega_0^2 \equiv k/m$,

$$\begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} = \omega_0^2 \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \qquad \text{i.e.} \qquad \ddot{\vec{x}} = \mathbf{M}\vec{x}. \tag{1.3}$$

The column is the *state* of the system (where both carts are), and $\mathbf{M}$ is a transformation acting on it. Intuitively, equations (1.1) and (1.2) just *look* like a matrix. More formally, we note that the

state we are transforming needs more than one number, and each derivative is a linear combination of those numbers.

Step 3: find the eigenvalues and eigenvectors of M. Both methods below start here, so let's get it out of the way. By definition $\mathbf{M}\vec{v} = \lambda\vec{v}$, so if we plug in $\vec{v}$, the right-hand side will simplify greatly. Let's see it in action. To find λ we need $\det(\mathbf{M} - \lambda\mathbf{I}) = 0$. We talked extensively in class about how this relates to linear dependence. We'll find λ and $\vec{v}$ for $\left(\begin{smallmatrix} -2 & 1 \\ 1 & -2 \end{smallmatrix}\right)$, knowing we can multiply by ω_0^2 at the end (if $\mathbf{N}\vec{v} = \lambda\vec{v}$ then $(\omega_0^2\mathbf{N})\vec{v} = \omega_0^2\lambda\vec{v}$: same eigenvectors, and the eigenvalues just pick up the factor):

$$\begin{vmatrix} -2-\lambda & 1 \\ 1 & -2-\lambda \end{vmatrix} = (-2-\lambda)^2 - 1 = \lambda^2 + 4\lambda + 3 = (\lambda+1)(\lambda+3) = 0$$
$$\Rightarrow \quad \lambda \in \{-1, -3\}. \tag{1.4}$$

Now find the eigenvectors: $(\mathbf{M} - \lambda\mathbf{I})\vec{v} = 0$ says

$$\begin{pmatrix} -2-\lambda & 1 \\ 1 & -2-\lambda \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad \text{i.e.} \qquad \begin{aligned} (-2-\lambda)x_1 + x_2 &= 0 \\ x_1 + (-2-\lambda)x_2 &= 0. \end{aligned} \tag{1.5}$$

Plug $\lambda = -1$ into both rows: $-x_1 + x_2 = 0$ and $x_1 - x_2 = 0$. The two rows say the same thing, as they must when the determinant is zero (that is exactly what $\det = 0$ buys you; if the rows disagreed, you'd have the wrong λ). So $x_2 = x_1$: only the ratio is fixed, and we choose $x_1 = 1$, or $\vec{v}_1 = \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$. Any multiple of an eigenvector is an eigenvector, so we could have picked $\left(\begin{smallmatrix} 2 \\ 2 \end{smallmatrix}\right)$, etc. Plug $\lambda = -3$ into both rows: $x_1 + x_2 = 0$ twice, so $\vec{v}_2 = \left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$. Putting the ω_0^2 back,

$$\boxed{\vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix},\ \lambda_1 = -\omega_0^2; \qquad \vec{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix},\ \lambda_2 = -3\omega_0^2.} \tag{1.6}$$

When we interpret this, we say "the eigenvalue is $-\omega^2$." Felder and Felder (p. 340) offer "the resonant frequencies of coupled oscillators are the square roots of the absolute values of the eigenvalues" as a tidbit worth keeping, and it is right here. But don't drop the sign: the rule works only because the eigenvalues are negative. The minus sign is the physics: a negative eigenvalue means oscillation; a positive one would mean exponential growth, and "the square root of its absolute value" would be a frequency for something that doesn't oscillate at all.

2 The first way: plug in one eigenvector at a time

$$\boxed{\vec{v}_1 = \begin{pmatrix}1\\1\end{pmatrix},\ \lambda_1 = -\omega_0^2; \qquad \vec{v}_2 = \begin{pmatrix}1\\-1\end{pmatrix},\ \lambda_2 = -3\omega_0^2.} \tag{1.6}$$

Each eigenvector becomes a normal mode. The first eigenvector tells us that if you plug in *any* state in which $x_2 = x_1$, the matrix will simply multiply that state by $-\omega_0^2$. Now, let's plug in. That means $x_1 = x_2$ at every time (so $\ddot{x}_2 = \ddot{x}_1$ as well), and Equation (1.3) becomes

$$\begin{pmatrix}\ddot{x}_1\\ \ddot{x}_2\end{pmatrix} = \omega_0^2 \begin{pmatrix}-2 & 1\\ 1 & -2\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix} \quad \rightarrow \quad \begin{pmatrix}\ddot{x}_1\\ \ddot{x}_1\end{pmatrix} = \omega_0^2 \begin{pmatrix}-2 & 1\\ 1 & -2\end{pmatrix}\begin{pmatrix}x_1\\ x_1\end{pmatrix}. \tag{2.1}$$

And now we use $\mathbf{M}\vec{v} = \lambda\vec{v}$: the column on the right is a multiple of the eigenvector $\vec{v}_1$, so $\mathbf{M}$ acting on it is just $\lambda_1 = -\omega_0^2$ times it, and we get

$$\begin{aligned}
\begin{pmatrix}\ddot{x}_1\\ \ddot{x}_1\end{pmatrix} &= \omega_0^2 \begin{pmatrix}-2 & 1\\ 1 & -2\end{pmatrix}\begin{pmatrix}x_1\\ x_1\end{pmatrix}\\
&= \mathbf{M}\ \vec{v}\\
&= \lambda\ \vec{v}\\
&= \omega_0^2\ (-1)\ \begin{pmatrix}x_1\\ x_1\end{pmatrix}\\
\rightarrow \quad \ddot{x}_1 &= -\omega_0^2\ x_1
\end{aligned} \tag{2.2}$$

(In fact we have this same equation $\ddot{x}_1 = -\omega_0^2 x_1$ twice.) The matrix multiplication became a scalar multiplication. That we know how to solve, and the solution is the first normal mode:

$$x_1(t) = A_1\cos\omega_0 t + A_2\sin\omega_0 t, \qquad x_2(t) = x_1(t). \tag{2.3}$$

Now the other eigenvector, $x_2 = -x_1$. The same steps, with $\left(\begin{smallmatrix}-2 & 1\\ 1 & -2\end{smallmatrix}\right)\left(\begin{smallmatrix}x_1\\ -x_1\end{smallmatrix}\right) = (-3)\left(\begin{smallmatrix}x_1\\ -x_1\end{smallmatrix}\right)$:

$$\begin{aligned}
\begin{pmatrix}\ddot{x}_1\\ -\ddot{x}_1\end{pmatrix} = -3\omega_0^2\begin{pmatrix}x_1\\ -x_1\end{pmatrix} \quad &\rightarrow \quad \ddot{x}_1 = -3\omega_0^2 x_1, \quad x_2 = -x_1\\
&\rightarrow \quad x_1(t) = B_1\cos\sqrt{3}\,\omega_0 t + B_2\sin\sqrt{3}\,\omega_0 t, \quad x_2(t) = -x_1(t).
\end{aligned} \tag{2.4}$$

Cool! That's the $\sqrt{3}$ we pulled out of a *hat* back when we first met these carts.[1]

The general solution is the sum of the two normal modes. Why are we allowed to add them? Equation (1.3) is linear, so if two motions solve it, their sum does too (superposition), even though the sum is no longer a multiple of either eigenvector. Two second-order equations need four constants, and the sum has exactly four:

$$\begin{aligned}
x_1(t) &= A_1\cos\omega_0 t + A_2\sin\omega_0 t + B_1\cos\sqrt{3}\,\omega_0 t + B_2\sin\sqrt{3}\,\omega_0 t\\
x_2(t) &= A_1\cos\omega_0 t + A_2\sin\omega_0 t - B_1\cos\sqrt{3}\,\omega_0 t - B_2\sin\sqrt{3}\,\omega_0 t.
\end{aligned} \tag{2.5}$$

Now we apply our initial conditions. Differentiate and set $t = 0$: $\dot{x}_1(0) = \omega_0 A_2 + \sqrt{3}\,\omega_0 B_2 = 0$ and $\dot{x}_2(0) = \omega_0 A_2 - \sqrt{3}\,\omega_0 B_2 = 0$, so $A_2 = B_2 = 0$. We discard the sine terms, and we're left with

$$\begin{aligned}
x_1(t) &= A_1\cos\omega_0 t + B_1\cos\sqrt{3}\,\omega_0 t\\
x_2(t) &= A_1\cos\omega_0 t - B_1\cos\sqrt{3}\,\omega_0 t.
\end{aligned} \tag{2.6}$$

[1] Try replacing the state with something that isn't a normal mode, say $x_2 = 3x_1$: the two rows give you two contradictory differential equations. Only the eigenvectors work!

Let's now be a little clever!

$$\begin{array}{c} \\ x_1(0) \\ x_2(0) \end{array} \begin{array}{c} \begin{array}{cc} A_1 & B_1 \end{array} \\ \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \end{array} \tag{2.7}$$

is the matrix that converts normal-mode amplitudes into initial positions. We can tell that from the column labels, and apparently it's a matrix of eigenvectors. Why? Well,

$$\begin{aligned} x_1 &= \overbrace{A_1 \cos\omega_0 t}^{\text{normal mode 1}} + \overbrace{B_1 \cos\sqrt{3}\,\omega_0 t}^{\text{normal mode 2}} \\ x_2 &= A_1 \cos\omega_0 t - B_1 \cos\sqrt{3}\,\omega_0 t \end{aligned} \tag{2.8}$$

Now plug in $t = 0$. The cosines are all 1, and we get

$$\begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} A_1 + B_1 \\ A_1 - B_1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} A_1 \\ B_1 \end{pmatrix}. \tag{2.9}$$

Each column is one normal mode at $t = 0$, which is an eigenvector. Cool. So, the matrix that converts normal-mode amplitudes to positions is the matrix of eigenvectors, and the matrix that converts positions to amplitudes is its inverse, with the labels swapped:

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}. \tag{2.10}$$

With its labels, which are those of (2.7) swapped,

$$\begin{array}{c} \\ A_1 \\ B_1 \end{array} \begin{array}{c} \begin{array}{cc} x_1(0) & x_2(0) \end{array} \\ \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \end{array}$$

converts initial conditions to normal-mode amplitudes. Applying it to ours,

$$\begin{pmatrix} A_1 \\ B_1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 3/2 \\ 3/2 \end{pmatrix}, \tag{2.11}$$

and the solution is

$$\boxed{x_1(t) = \tfrac{3}{2}\cos\omega_0 t + \tfrac{3}{2}\cos\sqrt{3}\,\omega_0 t, \qquad x_2(t) = \tfrac{3}{2}\cos\omega_0 t - \tfrac{3}{2}\cos\sqrt{3}\,\omega_0 t} \tag{2.12}$$

in cm, with $\omega_0 = \sqrt{k/m}$. **Whew!**

Check it at $t = 0$: $x_1 = 3$, $x_2 = 0$. Good. Pulling one cart and holding the other at rest is half of each normal mode, so each cart carries both frequencies, ω_0 and $\sqrt{3}\,\omega_0$. Their periods are in the ratio $\sqrt{3}$, whatever k and m are. And since two cosines with frequencies in an irrational ratio never line up again, the motion never exactly repeats. That's the "janky"[2] run we saw on the track.

One caution: the matrix in (2.7) is the matrix of eigenvectors *only at* $t = 0$. At a general t the matrix between (A_1, B_1) and (x_1, x_2) is

$$\begin{array}{c} \\ x_1(t) \\ x_2(t) \end{array} \begin{array}{c} \begin{array}{cc} A_1 & B_1 \end{array} \\ \begin{pmatrix} \cos\omega_0 t & \cos\sqrt{3}\,\omega_0 t \\ \cos\omega_0 t & -\cos\sqrt{3}\,\omega_0 t \end{pmatrix} \end{array}.$$

Setting $t = 0$ is what turned it into $\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$.[3]

[2] I will defend this as a technical term to my death.

[3] Is the general-t matrix always a scalar multiple of $\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$? No. It is the eigenvector matrix times a diagonal

3 The "better" way: the natural basis

$$\boxed{\vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix},\ \lambda_1 = -\omega_0^2; \qquad \vec{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix},\ \lambda_2 = -3\omega_0^2.} \tag{1.6}$$

This way gets the same answer, but it shows what "basis" and "diagonalization" mean more directly ... and it generalizes super well, as we'll see. Above, we plugged the eigenvectors in one at a time and watched what they did to the equation. Instead, let's rewrite the *entire* equation in the basis defined by the eigenvectors.

Question: Find the general solution to the same equations, the carts with equal masses m,

$$\begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} = \omega_0^2 \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

Answer: The eigenvectors of this transformation matrix are $\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$, with corresponding eigenvalues $-\omega_0^2$ and $-3\omega_0^2$ (boxed above). So, we can write x_1 and x_2 in this basis as

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = f(t) \begin{pmatrix} 1 \\ 1 \end{pmatrix} + g(t) \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{3.1}$$

This says "you can express the vector $\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right)$ at any given time as a linear combination of these two basis vectors, with f and g as the coefficients": f is how much of the first normal mode we have, g how much of the second. This changes in time, so f and g are functions of t. Instead of asking "where is each cart?" we are asking "how much of each normal mode do we have?"

Can we do that? Yes, for absolutely any functions x_1 and x_2, and not because $\vec{v}_1$ and $\vec{v}_2$ are eigenvectors: any two non-parallel vectors form a basis, so any vector is some combination of them. That's Section 6.4. Though our life will certainly be better here using an eigenbasis.

What happens when we do that? When we express all our states in a different basis, the entire equation changes, including the transformation matrix itself. We'd like to write a corresponding equation to (1.3) in this basis:

$$\begin{pmatrix} \ddot{f} \\ \ddot{g} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} f \\ g \end{pmatrix}, \tag{3.2}$$

and we need to know what a, b, c, d are. Two key points from linear algebra:

1. When you use the eigenvectors as a basis, the transformation is represented by a diagonal matrix of the eigenvalues. Let's write this out: call the eigenvectors $\vec{v}_1$, $\vec{v}_2$ with corresponding

matrix of the two mode factors,

$$\begin{pmatrix} \cos\omega_0 t & \cos\sqrt{3}\,\omega_0 t \\ \cos\omega_0 t & -\cos\sqrt{3}\,\omega_0 t \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \cos\omega_0 t & 0 \\ 0 & \cos\sqrt{3}\,\omega_0 t \end{pmatrix}.$$

Each column is still a multiple of its eigenvector (column 1 is $\vec{v}_1$ times $\cos\omega_0 t$, column 2 is $\vec{v}_2$ times $\cos\sqrt{3}\,\omega_0 t$), but the two columns are scaled by different amounts, so the whole matrix is a multiple of $\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ only when the two cosines are equal: at $t = 0$, and at isolated later instants, never over an interval, because the two frequencies differ. Two consequences: first, whenever either cosine is zero the determinant $-2\cos\omega_0 t\cos\sqrt{3}\,\omega_0 t$ vanishes, so at those instants you cannot recover the amplitudes from the positions alone (at every other instant you can, but the matrix you invert is not the eigenvector matrix; it equals that only at $t = 0$, where both cosines are 1). Second, the factorization is the whole story of the "better" way in §3: start in amplitude coordinates, let each mode evolve by its own scalar factor (the diagonal matrix), then convert to positions with the eigenvector matrix.

eigenvalues λ_1, λ_2, and take any vector $\vec{u} = \alpha\vec{v}_1 + \beta\vec{v}_2$. Apply $\mathbf{M}$ (the argument works for any matrix):

$$\mathbf{M}\vec{u} = \mathbf{M}(\alpha\vec{v}_1 + \beta\vec{v}_2) = \alpha\,\mathbf{M}\vec{v}_1 + \beta\,\mathbf{M}\vec{v}_2 = \alpha\lambda_1\vec{v}_1 + \beta\lambda_2\vec{v}_2.$$

Now read that in the $\{\vec{v}_1, \vec{v}_2\}$ basis. The input coordinates are $\left(\begin{smallmatrix}\alpha\\ \beta\end{smallmatrix}\right)$. The output is $\left(\begin{smallmatrix}\lambda_1\alpha\\ \lambda_2\beta\end{smallmatrix}\right)$. So

$$\mathbf{M} = \begin{pmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{pmatrix}$$

in this basis: the a, b, c, d of (3.2) are λ_1, 0, 0, λ_2. (§4 says the same thing more slowly.)

2. This naturally decouples our equations.

We can immediately write it. In the eigenvector basis, (1.3) becomes

$$\begin{pmatrix}\ddot{f}\\ \ddot{g}\end{pmatrix} = \begin{pmatrix}-\omega_0^2 & 0\\ 0 & -3\omega_0^2\end{pmatrix}\begin{pmatrix}f\\ g\end{pmatrix} \qquad \Rightarrow \qquad \ddot{f} = -\omega_0^2 f, \quad \ddot{g} = -3\omega_0^2 g.^{4} \tag{3.3}$$

The natural basis has done its job. Those are two separate simple harmonic oscillators, so

$$f = A_1\cos\omega_0 t + A_2\sin\omega_0 t, \qquad g = B_1\cos\sqrt{3}\,\omega_0 t + B_2\sin\sqrt{3}\,\omega_0 t. \tag{3.4}$$

Finally, we go back to our position basis (x_1, x_2) with (3.1), $x_1 = f + g$ and $x_2 = f - g$:

$$\boxed{\begin{aligned} x_1 &= f + g = A_1\cos\omega_0 t + A_2\sin\omega_0 t + B_1\cos\sqrt{3}\,\omega_0 t + B_2\sin\sqrt{3}\,\omega_0 t\\ x_2 &= f - g = A_1\cos\omega_0 t + A_2\sin\omega_0 t - B_1\cos\sqrt{3}\,\omega_0 t - B_2\sin\sqrt{3}\,\omega_0 t\end{aligned}} \tag{3.5}$$

So, four constants, as there must be, and it is exactly (2.5): the same answer as the first way.

The initial conditions, this way. Solving (3.1) for the coefficients gives $f = (x_1 + x_2)/2$ and $g = (x_1 - x_2)/2$ (that's the inverse matrix from §2 again, read as two equations). For $x_1(0) = 3$, $x_2(0) = 0$ that's $f(0) = g(0) = 3/2$: the same amplitudes, and the same solution (2.12). Physically, f is how far the midpoint of the two carts has moved from equilibrium, and g is half the compression of the middle spring, which is a nice way to see why the slow mode never stretches the middle spring and the fast one stretches it twice as much as either cart moves.

[4] You might ask: why isn't this matrix $\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$? Because that matrix is the converter, not the physics. Feed $\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$ the amounts of each mode, (f, g), and it returns positions, (x_1, x_2): that is (3.1) written as a matrix, and it is the matrix of eigenvectors, (2.7); call it $\mathbf{P}$. The matrix here takes the same input, (f, g), and returns accelerations, $(\ddot{f}, \ddot{g})$. Two checks. Dimensions: acceleration per displacement must carry an ω_0^2, and $\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$ is pure numbers. A pure mode: put in $f = 1$, $g = 0$; then $\vec{x} = \vec{v}_1$ and $\ddot{\vec{x}} = \mathbf{M}\vec{v}_1 = -\omega_0^2\vec{v}_1$, which in the $\{\vec{v}_1, \vec{v}_2\}$ basis is $(-\omega_0^2, 0)$, the first column above. If the matrix were $\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$ the first column would be $(1, 1)$, and a pure slow mode would accelerate into a mix of both modes. It doesn't, and that is the whole reason the matrix is diagonal. In Felder and Felder Section 7.2, you will learn more generally how to convert $\mathbf{M}$ from one basis to another, and you'll see that $\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$ appears on both sides of $\mathbf{M}$: the matrix in the new basis is $\mathbf{P}^{-1}\mathbf{M}\mathbf{P}$. You can check that here: $\mathbf{M}\mathbf{P} = \omega_0^2\left(\begin{smallmatrix}-2 & 1\\ 1 & -2\end{smallmatrix}\right)\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right) = \omega_0^2\left(\begin{smallmatrix}-1 & -3\\ -1 & 3\end{smallmatrix}\right) = \mathbf{P}\,\mathrm{diag}(-\omega_0^2, -3\omega_0^2)$, so $\mathbf{P}^{-1}\mathbf{M}\mathbf{P}$ is the diagonal matrix above.

4 Why is the matrix diagonal in the eigenvector basis?

Before we dive into the eigenproblem, let's review some general change-of-basis thoughts. The basis we're used to working with is the $\{\hat{\imath}, \hat{\jmath}\}$ basis: $\hat{\imath} = \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$, $\hat{\jmath} = \left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$. Let's pick another basis to consider, say $\vec{w}_1 = \hat{\imath} + \hat{\jmath}$ and $\vec{w}_2 = -3\hat{\imath} + 4\hat{\jmath}$. Then what we want is some matrix

$$\begin{array}{c} \\ \vec{w}_1 \\ \vec{w}_2 \end{array} \begin{array}{c} \begin{array}{cc} \hat{\imath} & \hat{\jmath} \end{array} \\ \begin{pmatrix} & \\ & \end{pmatrix} \end{array}$$

to turn a vector $\vec{u}$ written in the $\{\hat{\imath}, \hat{\jmath}\}$ basis into one written in the $\{\vec{w}_1, \vec{w}_2\}$ basis. The easy matrix to write down is

$$\begin{array}{c} \\ \hat{\imath} \\ \hat{\jmath} \end{array} \begin{array}{c} \begin{array}{cc} \vec{w}_1 & \vec{w}_2 \end{array} \\ \begin{pmatrix} 1 & -3 \\ 1 & 4 \end{pmatrix} \end{array}$$

whose columns are just $\vec{w}_1$ and $\vec{w}_2$. Now look at the labels: this is a matrix that eats a pair of $\{\vec{w}_1, \vec{w}_2\}$ amplitudes and spits out $\{\hat{\imath}, \hat{\jmath}\}$ amplitudes. (Feed it $\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$, meaning one $\vec{w}_1$ and no $\vec{w}_2$, and out comes $\left(\begin{smallmatrix}1\\1\end{smallmatrix}\right)$, which is $\vec{w}_1$ in $\hat{\imath}, \hat{\jmath}$ components.) That's the opposite of what we asked for. Its inverse,

$$\begin{pmatrix} 1 & -3 \\ 1 & 4 \end{pmatrix}^{-1} = \frac{1}{7}\begin{pmatrix} 4 & 3 \\ -1 & 1 \end{pmatrix},$$

does the opposite, which is the thing we wanted. Let's check: $\left(\begin{smallmatrix}2\\3\end{smallmatrix}\right)$ in $\{\hat{\imath}, \hat{\jmath}\}$ should go to some pair of $\{\vec{w}_1, \vec{w}_2\}$ amplitudes:

$$\frac{1}{7}\begin{pmatrix} 4 & 3 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \frac{1}{7}\begin{pmatrix} 8+9 \\ -2+3 \end{pmatrix} = \begin{pmatrix} 17/7 \\ 1/7 \end{pmatrix},$$

Now, if we've done things correctly, we can take those amplitudes times $\{\vec{w}_1, \vec{w}_2\}$ and recover our original vector $\left(\begin{smallmatrix}2\\3\end{smallmatrix}\right)$:

$$\frac{17}{7}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{7}\begin{pmatrix} -3 \\ 4 \end{pmatrix} = \begin{pmatrix} (17-3)/7 \\ (17+4)/7 \end{pmatrix} = \begin{pmatrix} 14/7 \\ 21/7 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.\checkmark$$

So: the matrix whose columns are the new basis vectors is the easy one to write, and it converts new-basis amplitudes into old-basis components. Going the other way means finding an inverse. That's exactly what happened with the matrix of eigenvectors, (2.7), in §2.

Now back to the eigenproblem. Felder and Felder say "This fact comes very directly from the definition of an eigenvector" and leave it there (p. 339; it's their Problem 6.206). We wrote it out under key point 1 in §3; here it is again, more slowly. The only facts we need are that matrix multiplication is linear, the definition $\mathbf{M}\vec{v}_i = \lambda_i \vec{v}_i$, and that $\vec{v}_1$ and $\vec{v}_2$ are independent, so they form a basis (true here, since the eigenvalues differ).

Let $\vec{v}_1$, $\vec{v}_2$ be the eigenvectors of $\mathbf{M}$ (any matrix would do), with eigenvalues λ_1, λ_2. Take any vector and write it in the eigenvector basis:

$$\vec{u} = \alpha\vec{v}_1 + \beta\vec{v}_2, \tag{4.1}$$

so $\vec{u}$ has coordinates (α, β) in the $\{\vec{v}_1, \vec{v}_2\}$ basis. Apply $\mathbf{M}$. It acts on each piece separately (linearity), and by the definition of an eigenvector it does nothing to each piece except scale it:

$$\mathbf{M}\vec{u} = \mathbf{M}(\alpha\vec{v}_1 + \beta\vec{v}_2) = \alpha\,\mathbf{M}\vec{v}_1 + \beta\,\mathbf{M}\vec{v}_2 = \alpha\lambda_1\vec{v}_1 + \beta\lambda_2\vec{v}_2. \tag{4.2}$$

Now read that in the $\{\vec{v}_1, \vec{v}_2\}$ basis. The input had coordinates $\binom{\alpha}{\beta}$. The output has coordinates $\binom{\lambda_1\alpha}{\lambda_2\beta}$. The only matrix that takes (α, β) to $(\lambda_1\alpha, \lambda_2\beta)$ is

$$\boxed{\mathbf{M} \text{ in the } \{\vec{v}_1, \vec{v}_2\} \text{ basis} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}} \tag{4.3}$$

That's the whole proof. In the $\{\hat{\imath}, \hat{\jmath}\}$ basis, $\mathbf{M}$ mixes the components (the off-diagonal entries are nonzero), because $\hat{\imath}$ and $\hat{\jmath}$ are not eigenvectors. In the eigenvector basis there is no mixing left to describe. Said another way: the columns of a matrix are the images of the basis vectors, and the image of $\vec{v}_1$ is $\lambda_1\vec{v}_1$, which in this basis is $(\lambda_1, 0)$.

For the springs, take $\vec{u}$ to be the state: plug (3.1) into (1.3) to get $\ddot{f}\vec{v}_1 + \ddot{g}\vec{v}_2 = f\lambda_1\vec{v}_1 + g\lambda_2\vec{v}_2$, and since $\vec{v}_1$ and $\vec{v}_2$ are independent, match coefficients: $\ddot{f} = \lambda_1 f$, $\ddot{g} = \lambda_2 g$. That is (3.3).

5 Which way should I use?

For one 2×2 problem by hand: it doesn't matter. The time-consuming part, finding the eigenvalues and eigenvectors (the determinant, the quadratic, two little linear systems), is shared, and neither method avoids it. After that, the first way substitutes each eigenvector in and fits the initial conditions with one 2×2 solve; the second way gets the decoupled equations in one line and fits the initial conditions with the same 2×2 solve dressed up as an inverse. The first way is a shade quicker (the substitution doubles as a check that your eigenvector is right); the second is a shade cleaner. Meh. A wash.

However, once the problem gets bigger (or gets reused), the second way wins:

- **Many degrees of freedom.** For N carts, both ways find N eigenvectors and fit N initial conditions; the second way packages all of it as one matrix product. On a computer that is the whole story: `numpy.linalg.eig` once, then, for carts released from rest, $\vec{x}(t) = \mathbf{P}\,\mathrm{diag}(\cos\omega_i t)\,\mathbf{P}^{-1}\vec{x}(0)$, where $\mathbf{P}$ is the matrix of eigenvectors as columns (the matrix in (2.7)) and $\omega_i = \sqrt{-\lambda_i}$. (Nonzero initial velocities add a second term of the same shape, with $\sin(\omega_i t)/\omega_i$ in place of $\cos\omega_i t$ and $\dot{\vec{x}}(0)$ in place of $\vec{x}(0)$.)

- **Reuse.** A driving force, or damping that is the same on every cart, leaves the eigenvectors alone. The second way then just adds a term to each one-dimensional equation; the first way has to redo the substitution. (New initial conditions cost the same either way: one 2×2 solve.)

- **Understanding.** The second way is the one that says *why* only those two ratios work: only in the basis of the normal modes does the matrix become diagonal; no other basis decouples the equations.

Felder and Felder (p. 340) put it this way. In both approaches the takeaway is "the eigenvectors of the matrix define the normal modes of the system." The second approach "is less direct, but it gives you perhaps the clearest view of why these normal modes work and no others would." My own one-line version: the first way is how you compute a given problem; the second is why the computation works. And their advice: "If you find the second approach confusing start by getting comfortable with the first approach ... but once you have worked with it a while come back to the second approach and see if you can see how it is doing the same thing as the first one, but in a different basis."

6 The general procedure, and the unequal-mass carts

Now for the capstone problem: you have the carts at the front of the room. And you can add masses to each of them. Your goal is to pick a set of masses, to predict the normal modes, and then to physically test it all out. To be clear, the three springs have equal spring constants k, but now your carts will have different masses. The general procedure is:

1. Draw the picture.
2. Write down N2L for each cart.
3. Turn it into a matrix equation $\ddot{\vec{x}} = \mathbf{M}\vec{x}$.
4. Find the normal modes: the eigenvectors of $\mathbf{M}$ give the ratio of x_2 to x_1 in each mode, and the eigenvalues give $-\omega^2$.
5. Combine them (either way) and use the initial conditions.

You can just jump in. If you want a small hint, here's how I dealt with notation: For different masses m_1 and m_2, it's convenient to just think about which mass is bigger and by how much. So, call the mass ratio $\mu = m_1/m_2$. Then, once you've gotten $\mathbf{M}$, you can pull out a k/m_1 so that it just looks like constants and μ.

Choosing your masses. Decide which cart is cart 1 before you write N2L (swapping the labels sends μ to $1/\mu$), and make the heavier cart about one and a half to three times the lighter one. Much closer to equal and the carts behave like the equal-mass case you've already solved. Much more unequal and the modes separate into "the light cart bounces while the heavy one barely moves" and "the heavy cart drags the light one along," which is easy to produce but hard to judge: almost any pluck of the light cart will look like a clean mode, because the heavy cart's response is too small to see.

How to tell you've isolated a mode. In a normal mode both carts turn around at the same instants, keep a fixed ratio of amplitudes, and nothing changes from cycle to cycle. If you missed, the energy sloshes: one cart's swing shrinks while the other's grows, and for a moment one cart sits nearly still while the other moves. Predict, before you touch the track, the sign of x_2/x_1 in each mode, roughly how big it is, and how many fast cycles fit in one slow cycle; then count, don't time.

Here's my solution: step 2 gives $m_1\ddot{x}_1 = -2kx_1 + kx_2$ and $m_2\ddot{x}_2 = kx_1 - 2kx_2$. Divide each line by its own mass:

$$\begin{aligned} \ddot{x}_1 &= -2\frac{k}{m_1}x_1 \quad + \frac{k}{m_1}x_2 \\ \ddot{x}_2 &= \quad \frac{k}{m_2}x_1 - 2\frac{k}{m_2}x_2 \end{aligned}$$

which is already shaped like a matrix equation. Now put in μ: since $k/m_2 = \mu\,(k/m_1)$,

$$\begin{aligned} \ddot{x}_1 &= -2\frac{k}{m_1}x_1 \quad + \frac{k}{m_1}x_2 \\ \ddot{x}_2 &= \quad \mu\frac{k}{m_1}x_1 - 2\mu\frac{k}{m_1}x_2 \end{aligned}$$

and pulling the common k/m_1 out front,

$$\mathbf{M} = \frac{k}{m_1}\begin{pmatrix} -2 & 1 \\ \mu & -2\mu \end{pmatrix}. \tag{6.1}$$

Plug in your μ and do step 4. The k/m_1 out front just scales every eigenvalue, so write $\lambda = (k/m_1)\,\ell$ and find the eigenvalues ℓ of the matrix in the parentheses, call it $\mathbf{M}'$:

$$0 = \det\begin{pmatrix} -2-\ell & 1 \\ \mu & -2\mu-\ell \end{pmatrix} = (-2-\ell)(-2\mu-\ell) - \mu = \ell^2 + 2(1+\mu)\,\ell + 4\mu - \mu,$$

$$0 = \ell^2 + 2(1+\mu)\,\ell + 3\mu \qquad \Longrightarrow \qquad \ell = -(1+\mu) \pm \sqrt{(1+\mu)^2 - 3\mu} = -(1+\mu) \pm \sqrt{\mu^2 - \mu + 1}.$$

Call that square root $s = \sqrt{\mu^2 - \mu + 1}$ for short. For the eigenvectors, the top row of $(\mathbf{M}' - \ell\,\mathbf{I})\vec{v} = 0$ says $(-2-\ell)x_1 + x_2 = 0$, so $x_2 = (2+\ell)\,x_1 = (1 - \mu \pm s)\,x_1$; take $x_1 = 1$. (The bottom row says the same thing, as it must.)

So the eigenvectors are no longer $\left(\begin{smallmatrix}1\\1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1\\-1\end{smallmatrix}\right)$; they are

$$\vec{v}_{\text{slow}} = \begin{pmatrix} 1 \\ 1-\mu+s \end{pmatrix} \qquad \text{and} \qquad \vec{v}_{\text{fast}} = \begin{pmatrix} 1 \\ 1-\mu-s \end{pmatrix}, \tag{6.2}$$

with corresponding eigenvalues $\lambda = (k/m_1)\,\ell$,

$$\lambda_{\text{slow}} = \frac{k}{m_1}\big(-(1+\mu)+s\big) \qquad \text{and} \qquad \lambda_{\text{fast}} = \frac{k}{m_1}\big(-(1+\mu)-s\big). \tag{6.3}$$

(Check: $\mu = 1$ gives $s = 1$, so $\lambda = -k/m_1$ and $-3k/m_1$ with $x_2/x_1 = \pm 1$, as in (1.6).) The heavier cart moves more in the slow mode and less in the fast one. The ratio of the two periods should turn out to be $T_{\text{slow}}/T_{\text{fast}} = \sqrt{\lambda_{\text{fast}}/\lambda_{\text{slow}}}$.